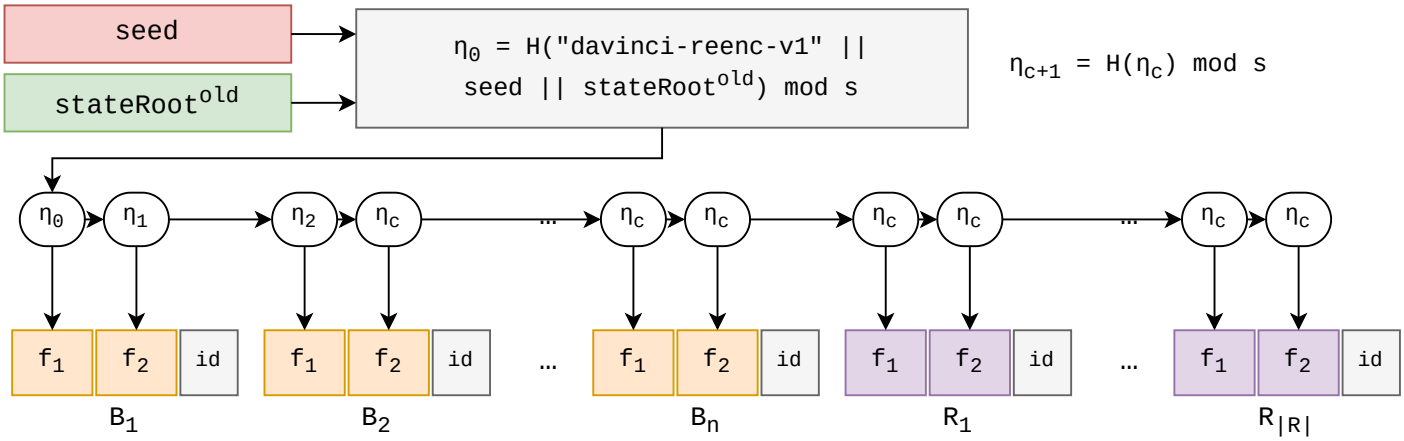

Each active ciphertext consumes the next scalar η_c; inactive fields (id) must be the identity and consume none (example with $n_f = 2$).

re-encryption with η_c:

$(C_1, C_2) \rightarrow (C_1 + \eta_c \cdot G,$
$C_2 + \eta_c \cdot \text{encryptionKey})$

no scalar repeats within or across batches

results accumulation

$A^{new} = A^{old} + \Sigma\ B'_i - \Sigma\ B^{old}{}_i$
$+ \Sigma\ (R'_j - R_j)$

refreshes add encryptions of zero